Project Title

Virtual Memory Page Replacement Visualizer

Project Report submitted for the partial fulfillment of grade for the course

Operating Systems And Systems Programming (25SC2104E)

in

A.Y. 2026-2027

Submitted by

SAI SURENDRANATH PENDYALA (2520030157)
ADA ROHAN (2520030195)
MEDIKONDA RAM RATAN (2520030291)

Team Number: 14
Section: 4

Under the guidance of

Mrs. E. SOUMYA

Asst. Professor, CSE



Hyderabad-500090, Telangana, India.

ABSTRACT

The Virtual Memory Page Replacement Visualizer is a web-based application designed to demonstrate and compare three page replacement algorithms used in operating systems: First In, First Out (FIFO), Least Recently Used (LRU), and Optimal Page Replacement. When physical memory becomes full and a new page needs to be loaded, the operating system must select a page to replace. The proposed system simulates this process using a user-defined page reference string and a specified number of memory frames. It visually displays the contents of memory frames after each page request and identifies page hits and page faults. The system calculates the total number of page faults generated by each algorithm and presents a comparative analysis using tables and graphical representations. This helps users understand how different page replacement strategies operate and how their performance varies with different input sequences. The application can be developed using HTML, CSS, and JavaScript, providing an interactive and user-friendly interface for learning operating system memory management concepts. The proposed visualizer serves as an educational tool for students to understand virtual memory, page replacement techniques, and the importance of reducing page faults to improve memory management efficiency.

Signature of the Guide